

KARAIKAL, India

WMO: 433460

Lat: 10.917N

Lon: 79.833E

Elev: 23

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.4	72.7	64.3	90.3	79.3	65.7	94.8	79.6	18.5	79.7	16.7	80.0	4.6	320	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.3	99.0	79.9	97.6	79.8	96.2	79.9	85.3	92.5	84.5	91.9	83.7	91.1	8.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.5	175.5	89.9	82.6	170.3	89.5	81.8	165.7	88.8	49.8	92.8	48.9	92.1	47.9	91.3	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	14.3	13.0	DB	68.3	102.5	1.0	1.5	67.6	103.6	67.0	104.5	66.5	105.4	65.8	106.5	(4)
			WB	65.3	88.1	1.2	2.4	64.4	89.8	63.7	91.2	63.1	92.6	62.2	94.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.2	78.7	80.4	83.5	86.8	88.9	88.7	88.1	86.6	85.6	83.5	80.4	79.1	
	DBStd	4.27	1.82	2.03	2.08	2.03	2.68	2.30	1.99	2.45	2.00	2.75	2.14	2.10	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	12484	889	850	1037	1104	1206	1161	1182	1135	1069	1037	912	902	
	CDD65	7009	424	430	572	654	741	711	717	670	619	572	462	437	
	CDH74	85080	3550	4207	6884	9000	10343	9780	9692	8657	7814	6684	4649	3820	
	CDH80	38173	537	1062	2743	4730	5934	5506	5275	4364	3656	2615	1155	596	
Wind	WSAvg	6.7	7.9	6.7	5.9	5.9	6.8	7.5	7.3	6.7	6.0	5.0	6.5	8.0	
Precipitation	PrecAvg	48.9	1.5	0.7	0.8	1.1	1.9	1.5	1.6	3.1	3.3	8.6	16.1	9.0	
	PrecMax	69.7	7.8	3.6	7.3	8.9	7.4	6.1	5.0	9.5	8.0	29.0	33.2	18.0	
	PrecMin	24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.4	0.0	0.0	0.7	1.6	
	PrecStd	11.4	1.7	1.1	1.7	1.9	1.9	1.8	1.2	2.1	1.9	6.0	8.1	6.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	89.1	93.0	97.6	101.6	100.8	99.5	98.6	96.8	93.9	89.3	86.4	
		MCWB	77.4	79.8	81.6	82.6	81.6	79.3	79.5	77.9	78.3	79.3	79.0	78.0	
	2%	DB	84.7	87.4	91.2	95.0	99.2	98.7	97.6	96.7	94.5	92.1	87.4	85.1	
		MCWB	76.3	78.7	80.8	83.2	81.6	79.0	78.7	78.2	78.9	79.5	79.1	77.7	
	5%	DB	83.8	86.2	90.0	93.4	97.2	97.0	96.2	94.9	93.0	90.6	86.1	84.2	
		MCWB	75.7	77.2	80.1	83.3	81.9	79.5	78.6	78.3	79.2	79.6	78.4	77.2	
	10%	DB	82.9	85.2	88.9	92.2	95.1	95.3	94.6	93.1	91.5	89.0	84.8	83.1	
		MCWB	75.2	76.5	79.5	82.8	82.1	79.7	78.6	78.5	79.5	79.6	78.0	76.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.5	81.6	84.1	86.7	87.0	85.4	84.8	84.3	84.5	82.7	81.8	80.5
			MCDB	83.9	87.4	90.7	92.9	94.9	92.9	92.8	90.8	89.2	89.6	86.3	84.0
2%		WB	78.3	79.9	82.3	85.2	85.3	84.1	83.4	82.4	82.8	81.6	80.6	79.4	
		MCDB	82.7	85.8	89.0	91.9	93.2	92.3	91.9	89.9	88.9	88.2	85.3	83.0	
5%		WB	77.4	78.8	81.2	84.3	84.5	83.0	82.2	81.5	81.8	80.9	79.8	78.7	
		MCDB	81.8	84.5	88.0	91.2	92.4	90.9	90.7	89.0	88.3	87.1	84.2	82.4	
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.1	11.2	11.6	14.3	15.2	14.7	14.4	13.1	10.4	8.3	7.9	
		MCDBR	9.4	10.4	12.1	12.8	17.1	17.1	16.5	16.1	15.0	12.8	9.8	8.6	
		MCWBR	4.9	4.9	5.3	5.3	6.3	6.5	6.4	6.1	6.1	5.3	4.4	4.6	
	5% WB	MCDBR	8.5	9.7	11.5	12.1	14.0	15.2	15.2	14.3	13.2	11.2	8.6	7.8	
		MCWBR	5.1	5.2	5.6	5.7	6.3	7.3	7.3	6.7	6.2	5.3	4.5	4.8	
	Clear-Sky Solar Irradiance	taub	0.455	0.448	0.483	0.499	0.583	0.518	0.526	0.516	0.505	0.523	0.504	0.495	
taud		2.247	2.262	2.166	2.178	1.945	2.153	2.123	2.160	2.193	2.135	2.163	2.149		
Ebn at Noon		270	277	268	261	235	248	247	253	258	252	253	254		
Edh at Noon		44	44	49	49	60	48	50	49	48	50	47	47		
All-Sky Solar Radiation	RadAvg	1593	1877	2030	2009	1909	1819	1766	1800	1795	1527	1305	1335		
	RadStd	83	80	94	95	119	108	95	89	107	97	138	112		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.50	N/A	N/A	+1.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page